

Vineeth Kumar Sanda

Austin, Texas | 512-786-7336 | kumarvineeth408@gmail.com | [Linkedin](#) | [Github](#) |

SUMMARY

Generative AI Engineer with proven expertise in Machine Learning and Deep Learning, specializing in AI solutions. Proficient in developing virtual assistants using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), enhancing customer service efficiency. Skilled in Natural Language Processing techniques to boost chatbot engagement and experienced with cloud integration, MongoDB Atlas, creating Webservice using FastAPI and hands on experience in CI/CD pipeline management using Azure DevOps.

EXPERIENCE

• Gen AI / ML Engineer

JAN 2024 – Present

Citizens Bank

Austin, Texas

- Collaborated with the data science team to generate and curate large-scale datasets for training and evaluating fine-tuned Llama3 and transformer-based models which enhances model robustness through data augmentation and cleaning techniques, ultimately improving overall performance in domain-specific use cases.
- Designed and developed an AI-powered virtual assistant leveraging Large Language Models (LLMs) and LangChain to streamline customer interactions, enhancing response accuracy and efficiency
- Developed Optical Character Recognition(OCR) pipelines using AWS Textract to extract and process text from scanned documents, improving data availability for downstream NLP and RAG tasks
- Utilized GPT-4o for data generation, including automatic summarization and content generation, ensuring the outputs were well-suited for RAG techniques.
- Integrated Retrieval-Augmented Generation (RAG) using Vector Databases (ChromaDB, Pinecone) to enable context-aware responses.
- Utilized MongoDB Atlas to manage conversation history and user profiles for real-time personalization.
- Conducted research on various Advanced RAG techniques including rerankers, metadata filtering with intent to reduce the search space, various prompt optimization techniques like CoT, Reflection and Guard-railing the outputs of LLM model to streamline the outputs for the specified use case.
- Built and Deployed FastAPI based endpoints to enable real-time model inference and interaction with LLMs .
- Collaborated with Devops and Cloud teams to ensure seamless deployments and robust scaling through dockerized solutions using Azure CI/CD pipelines.

• Machine Learning Engineer

SEP 2020 – JULY 2022

Nisum Consulting

Hyderabad, India

- Developed and Maintain (includes build, deploy, debug) REST APIs (Flask), Scheduled Services to provide recommendations to Apps which include textual Analysis, machine learning capabilities on agile scrum data.
- Designed and build machine learning and NLP solutions for predictive analysis on agile scrum data including user stories duplicate detection and classification of user stories for scrum master dashboard applications.
- Expanded machine learning capabilities leveraging Natural Language Processing (NLP) for over five languages, demonstrating proficiency in cross-linguistic NLP Techniques using Spacy and NLTK libraries support for multiple languages.
- Conducted extensive EDA and multiple Machine Learning algorithms(Logistic Regression) to build a model for predicting viability for a user story based on historical Data.
- Leveraged python frameworks like pycaret, mlflow to automates much of the machine learning pipeline, including training, tuning, and evaluation across multiple algorithms.
- Deployed ML solutions as web services using Azure based YAML files by collaborating with Devops/Cloud teams.

• Software Engineer Associate

June 2019 – Aug 2020

• *Nisum Consulting*

Hyderabad, India

- Designed and developed Python APIs for the Scrum Master dashboard, streamlining the analysis of agile data (user stories, features, test cases, bugs) to improve team productivity.
- Built and maintained automated Python services (Python ApSchedulers) for scheduled tasks such as data extraction and reporting, ensuring efficiency and reliability .
- Integrated APIs with internal platforms to enhance user experience, collaborating with cross-functional teams to translate business requirements into technical solutions.
- Deployed and maintained APIs as microservices on AWS EC2 instances, utilizing Nginx as a load balancer to ensure high availability and scalability.
- Optimized API performance, reducing response time by 30% and enhancing scalability to handle increasing data volume.
- Developed an automated deployment script to compile python code through the Azure pipeline that helped to

decrease the deployment time and supported offline deployments for on-prem clients.

PERSONAL PROJECTS

- Build a deep neural network (CNN) with a concept of Focal loss which tries to handle the class imbalance problem by assigning more weights to hard or easily misclassified examples. Able to achieve better accuracy than simple CNN.
- Fine-tuned and optimized the phi-4 (14B) model using unsloth and vLLM , implementing LoRA-based Adaptation for efficient inference.

TECHNICAL SKILLS

Tools/Technologies: Python, Java, C++, R, SQL, MongoDB ,NLP ,NLTK, Spacy, Scikit-Learn, BERT, LayoutLM, LLMs, Langchain, Pinecone, Open AI API, TensorFlow, Keras, PyTorch, Kafka, Rabbit MQ, Spark, Hadoop, Hive, AWS ML, Textract, Sage Maker, S3, EC2, ECR, CI/CD, A/B Testing, FAISS, Solr, Lucene, Flask, Docker, MLOPS,Kubernetes, ReactJS, A/B Testing, Prefect, Airflow, Statistics, Tableau, Hugging Face, Git, GitHub, Jira

EDUCATION

- **Lindsey Wilson College** Columbia ,Kentucky
Master's in computer science
- **Mahatma Gandhi Institute of Technology** Hyderabad,India
Bachelor's in computer science engineering